Lecture 24

Fundamentals of Numerical Analysis

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Review of Lecture 23: Approximate Solutions

- Direct solves (GE, LU) cost $\mathcal{O}(n^3)$ for a full system; we often only need $\hat{x}$ with $\|\hat{x} - x^*\| \leqslant \epsilon$.
- The *residual* $\hat{r} = b - A\hat{x}$ is computable and measures the error:

$$\hat{r} = A\hat{e}, \qquad \hat{e} = x^* - \hat{x}$$

- Ideal update $x^{(1)} = x^{(0)} + A^{-1}r^{(0)}$ needs A^{-1}, so replace it with a cheap $Q^{-1} \approx A^{-1}$:

$$x^{(k)} = x^{(k-1)} + Q^{-1}r^{(k-1)} \quad \Longleftrightarrow \quad x^{(k)} = Q^{-1}(Q - A)x^{(k-1)} + Q^{-1}b$$

- Two popular choices:
  - Jacobi: $Q = D = diag(A)$
  - Gauss-Seidel: $Q = D - L = lowertri(A)$

Review of Lecture 23: Convergence and Cost

- Error propagation:

$$e^{(k)} = (I - Q^{-1}A)^k e^{(0)}, \qquad \|e^{(k)}\| \leqslant \|I - Q^{-1}A\|^k \|e^{(0)}\|$$

  so the iteration converges when $\|I - Q^{-1}A\| < 1$.
- Sufficient condition: if A is *diagonally dominant*, $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all i, then Jacobi and Gauss-Seidel converge for any $x^{(0)}$.
- Cost: with m nonzeros per row, each sweep uses $\mathcal{O}(mn)$ operations.
- Jacobi stores updates in a new vector; Gauss-Seidel uses each updated $x_i^{(k)}$ immediately.
- Both can be viewed as a form of averaging.

Today: a different approach for symmetric positive definite A — Conjugate Gradients.

Quadratic Form

$$f(x) = \frac{1}{2}x^T Ax - b^T x + c$$

is a *quadratic* form. Example:

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ -8 \end{bmatrix} \quad c = 0$$

```
syms x y
X = [x;y];
A = [3 2; 2 6]; b= [2; -8];
f = (1/2) * transpose(X) * A * X - transpose(b) * X;

ezsurfc(f)
hold on;
ezplot('(1/2) * (2 - 3*x)');
ezplot('(1/6) * (-8 - 2*x)');
```

Quadratic Form

Gradient: $\nabla f(x)$ points uphill

```
df = [diff(f,x); diff(f,y)]
[xx,yy] = meshgrid(linspace(-6,6,20));
U=subs(df(1),{x,y},{xx,yy});
V=subs(df(2),{x,y},{xx,yy});
quiver(xx,yy,U,V);
```

Quadratic Form

Lets look at the ith component of ∇f:

$$\begin{aligned} f(x+he_i) &= \frac{1}{2}(x+he_i)^T A(x+he_i) - b^T(x+he_i) + c \\ &\approx \frac{1}{2}(x^T Ax + he_i^T Ax + x^T Ahe_i) - b^T(x+he_i) + c \end{aligned}$$

so

$$\begin{aligned} \frac{f(x+he_i) - f(x)}{h} &= \frac{\frac{1}{2}(he_i^T Ax + x^T Ahe_i) - he_i^T b}{h} \\ &= \text{i}^{th} \text{ component of } \frac{1}{2}(Ax + A^T x) - b \end{aligned}$$

So

$$\nabla f = \frac{1}{2}A^T x + \frac{1}{2}Ax - b$$

So if A is symmetric, then $Ax = b$ at the minimum of f.

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So if A is symmetric, then $Ax = b$ at the minimum of f. Now, if

A is positive definite, f is concave up.
A is negative definite, f is concave down.
A is positive semi-definite, f is concave up (with a line as the minimum).
A is indefinite, f has a saddle. (Think $x_1^2 - x_2^2$)

Conjugate Gradients

- Suppose that A is $n \times n$ symmetric and positive definite.
- Since A is positive definite, $x^T Ax > 0$ for all $x \in \mathbb{R}^n$.
- Define a quadratic function

$$f(x) = \frac{1}{2}x^T Ax - x^T b$$

- It turns out that $-\nabla f = b - Ax = r$
- Optimization methods look in a "search direction" and pick the best step:

$$x_{k+1} = x_k + \alpha s_k$$

  Choose α so that $f(x_k + \alpha s_k)$ is minimized in the direction of s_k.
- Find α so that f is minimized:

$$0 = \frac{d}{d\alpha} f(x_{k+1}) = \nabla f(x_{k+1})^T \frac{d}{d\alpha} x_{k+1} = -r_{k+1}^T \frac{d}{d\alpha}(x_k + \alpha s_k) = -r_{k+1}^T s_k.$$

Steepest Descent

- Find α so that f is minimized in the direction of $-\nabla f = r$:

$$0 = \frac{d}{d\alpha} f(x_{k+1}) = \nabla f(x_{k+1})^T \frac{d}{d\alpha} x_{k+1} = -r_{k+1}^T \frac{d}{d\alpha}(x_k + \alpha r_k) = -r_{k+1}^T r_k.$$

- Pick α so that $\nabla f(x_{k+1})$ and r_k are orthogonal.
- Since $\nabla f(x_{k+1}) = -r_{k+1}$ we want

$$\begin{aligned} r_{k+1}^T r_k &= 0 \\ (b - A(x_k + \alpha r_k))^T r_k &= 0 \\ (b - Ax_k)^T r_k - \alpha (Ar_k)^T r_k &= 0 \\ r_k^T r_k &= \alpha r_k^T A r_k \end{aligned}$$

- So, the optimal search parameter is

$$\alpha = \frac{r_k^T r_k}{r_k^T A r_k}$$

Conjugate Gradients

- So, the optimal search parameter is

$$\alpha = \frac{r_k^T s_k}{s_k^T A s_k}$$

- This is CG: take a step in a search direction

Congugate Gradients

- Neat trick: We can compute the r without explicitly forming $b - Ax$:

$$r_{k+1} = b - Ax_{k+1} = b - A(x_k + \alpha s_k) = b - Ax_k - \alpha As_k = r_k - \alpha As_k$$

Congugate Gradients

- How should we pick s_k?
- Note that $-\nabla f = b - Ax = r$, so r is the negative gradient of f (for *any* x), and this is a good direction.
- Thus, pick $s_0 = r_0 = b - Ax_0$.
- What is s_1? This should be in the direction of r_1, but *conjugate* to s_0: $s_1^T A s_0 = 0$.
- (Two vectors u and v are A-*conjugate* is $u^T A v = 0$)
- So, if we let $s_1 = r_1 + \beta s_0$, we can require

$$0 = s_1^T A s_0 = (r_1^T + \beta s_0^T) A s_0 = r_1^T A s_0 + \beta s_0^T A s_0$$

or

$$\beta = -r_1^T A s_0 / s_0^T A s_0.$$

- Holds for $s_{k+1} = r_{k+1} + \beta_{k+1} s_k$
- Further similification (which is *not* simple to carry out) yields a simple method that requires only one matrix-vector product per step:

Conjugate Gradients

$x_0 =$ `initial guess`
$r_0 = b - Ax_0$
$s_0 = r_0$
for $k = 0, 1, 2, \ldots$
    $\alpha_k = \frac{r_k^T r_k}{s_k^T A s_k}$
    $x_{k+1} = x_k + \alpha_k s_k$
    $r_{k+1} = r_k - \alpha_k A s_k$
    $\beta_{k+1} = r_{k+1}^T r_{k+1} / r_k^T r_k$
    $s_{k+1} = r_{k+1} + \beta_{k+1} s_k$

Homework: investigate speed/accuracy of CG versus Jacobi/GS

So far, we've *motivated* a method. We haven't been careful about the step direction s_k yet.

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- Better: step toward a principle axis
- Notice: axis defined by eigenvectors of A
- Compromise: find a step direction that is A-orthogonal to the previous
- Notice: take a search closest to r that is conjugate.
- Result: converges in at most n steps

Conjugate Gradients

1 $x_0 =$ `initial guess`
2 $r_0 = b - Ax_0$
3 $s_0 = r_0$
4 `for` $k = 0, 1, 2, \ldots$
5 $\quad \alpha_k = \frac{r_k^T r_k}{s_k^T A s_k}$
6 $\quad x_{k+1} = x_k + \alpha_k s_k$
7 $\quad r_{k+1} = r_k - \alpha_k A s_k$
8 $\quad \beta_{k+1} = r_{k+1}^T r_{k+1} / r_k^T r_k$
9 $\quad s_{k+1} = r_{k+1} + \beta_{k+1} s_k$

Convergence

- A-norm of the error is minimized in each step
- Bound:

$$\|e_k\|_A \leqslant 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|e_0\|_A$$

- κ is the condition number
- for symmetric matrices, $\kappa = \frac{\lambda_{max}}{\lambda_{min}}$

Preconditioning

We've seen iterations that replace A with a matrix Q such that $Q^{-1}A$ resembles the identity matrix. Congugate Gradient takes a different approach and solves $Ax = b$ more directly.
Can we combine these ideas?
Preconditioning is transforming the original problem $Ax = b$ with another matrix Q and then solving that modified problem. For example

- Left Preconditioning: $QAx = Qb$
- Right Preconditioning: $AQw = b$, $Qw = x$
- Symmetric preconditioning: $Q^TAQw = Q^Tb$, $Qw = x$

For Congugate Gradient, the matrix must be symmetric, so the last form is the most common.
There are methods that generalize conjugate gradient for nonsymmetric matrices.

Coming Up

- Google, Markov Chains, intro to Monte Carlo Simulations
- Eigenvalues: SVD, Power Method
- Least-Squares
- Monte Carlo Simulations

Google (basics)

- A component of Google's success can be attributed to the PageRank^{TM} algorithm developed by Google's founders
- algorithm determined entirely by link structure of the WWW
- recomputed once a month
- involves no content of any Web page
- How are the search results ordered?